

TAMPINES JUNIOR COLLEGE PRELIMINARY EXAMINATION 2008

BIOLOGY Paper 2 Core Paper

8875/02

QUESTION ONE

(a)	Secondary structure – alpha helices, beta-pleated sheets;	1
	Tertiary structure – 3-dimensional, globular shape;	1
	Quaternary structure - 2 subunits / 2 polypeptide chains;	1
(b)	(i) microtubule organising centre / MTOC; (also accept centrosome/ centriole/ nuclear envelope/ kinetochore)	1
	(ii) they are spindle fibres for separating sister chromatids / chromosomes / kinetochores;	1
(c)	The chromosomes attached to microtubules at centromeres would line up on the equator of the spindle and at right angles to the spindle axis;	1
	Centromeres of each chromosome divides;	1
	Sister chromatids separate and sister chromatids become daughter chromosomes;	1
	Daughter chromosomes move to opposite poles of the spindle as spindle fibres shorten so that the spindle fibres/ kinetochore attached to microtubules pull daughter centromeres and daughter chromosomes apart;	1
		Max. 3
(d)	The exact replication of DNA from the parental chromosomes before mitosis occurs;	1
	The arrangement of the chromosomes on the spindle ensures that the sister chromatids are distributed evenly between the 2 daughter cells;	1
	Mitosis ensures that the daughter cells receive precisely the same number and types of chromosomes as the original parent cell;	1
	No chiasmata formation and therefore no crossing over;	1
		Max. 3
	Total	Max. 11

QUESTION TWO



(a)	(i) transfer RNA	1
	(ii) hydrogen bond	1
(b)	codon	1
(c)	TTA	1
(d)	Endoplasmic reticulum – transports the protein	1
	Golgi apparatus – transport and modification of protein eg. Addition of COH	1
(e)	Cellulase Structure – globular Function – enzymatic	1 1
	Collagen Structure - fibrous	1
	Total	Max. 9

QUESTION THREE



The following figure shows the main stages in the polymerase chain reaction (PCR).

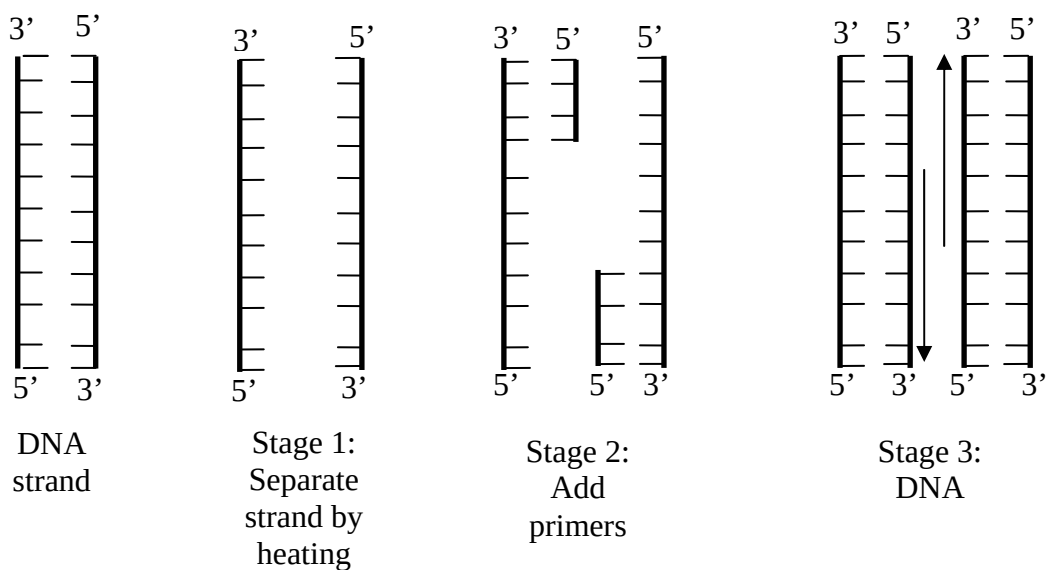


Fig. 4.1

(a) With reference to Fig. 4.1, explain what is happening at each of the three stages.

(i) Stage 1:

**Denaturates/H bonds broken;
2 DNA strands separate;
 by heating to 95°C; (A) 90-100°C**

[2]

(ii) Stage 2:

**Primers/oligonucleotides bind to, single DNA/strand/3' ends;
 Annealing/formation of H bonds;
 By cooling to 54°C; (A) 30-65°C**

[2]

(iii) Stage 3:

**New strands (of DNA) are synthesized by, Taq DNA polymerase;
 Starts at the position of primers;
 Addition of nucleotides to the 3' ends/ elongation takes place in a 5' to 3' direction;
 By heating to 72°C; (A) 60-75°C**

[2]



- (b) Suggest one advantage and one disadvantage of using PCR.

Advantage:

Allow amplification of minute quantities of DNA sample;
For easier analysis in forensic science, DNA fingerprinting;
Sensitive;

[2]

Disadvantage:

Taq polymerase lacks 3'- 5' proofreading mechanism;
Accumulation of error may occur;

- (c) Explain the advantages of using Taq DNA polymerase.

Not denatured at high temperatures;

Allows automation of PCR as fresh DNA polymerase does not need to be added after each cycle;



QUESTION FOUR

(a)	Conditions: low levels of oxygen;	1
	Abnormal beta chains make the haemoglobin molecules less soluble, tend to stick together, forming long fibres inside the red blood cells.	1
	Ref. to the change in structure and shape of the haemoglobin, mentioning the change in the bonding and the hydrophobic groups facing outwards;	1
	Glutamic acid is hydrophilic in nature, valine is hydrophobic in nature => hence ionic cross link is not formed and hence there is a change in the shape of the tertiary structure of haemoglobin;	
	The red blood cells are pulled out of shape, into a crescent or sickle shape.	1
	Bonus mark allowed	Max. 3
(b)	Probability: 1 in 4/ $\frac{1}{4}$ / 0.25 / 25%	1
	Derivation:	
	Parental Phenotypes: Heterozygous for sickle cell anaemia for both	
	Parental Genotypes: Hb^A Hb^S X Hb^A Hb^S	1
	Gametes: Hb^A Hb^S Hb^A Hb^S	1
	Offspring Genotypes: Hb^A Hb^A Hb^A Hb^S Hb^A Hb^S Hb^S Hb^S	1
(c)	Offspring Phenotypes: Normal Heterozygous for Sick cell anaemia Sick cell anaemia	1
	Distribution same as / very similar to, the malarial parasite;	1
	Resistant to malaria;	1
	Some migration into Nairobi;	1
	Allele carried by, immigrants/ urban commuters;	1
	Similar to breeding grounds of the mosquito;	1
	Act as a vector/ carries malarial parasite;	1
	(sickle cell) not found above 1 500 m;	1
	No malaria/ no mosquitoes/ low oxygen;	1
(d)		Max. 2
	Heterozygote population / proportion of sickle cell allele / gene frequency falls;	1
	No longer of selective advantage to be resistant to malaria / normal homozygous genotype has no selective disadvantage;	1
	Instead is now at a selective disadvantage to be suffering from anaemia / oxygen deprivation;	1
	Bonus mark allowed	Max. 3
	Total	Max. 10



QUESTION FIVE

- (a) Describe features of a plasmid that made it suitable as a vector. [4]
- (b) Using a named example, outline the process of cloning a eukaryotic gene into *E.coli* cells to produce eukaryotic protein in genetic engineering. [10]
- (c) Discuss the goals and implications of the human genome project, including the benefits and difficult ethical concerns. [6]

- (a) Describe features of a plasmid that made it suitable as a vector.
You may want to illustrate using a drawing of a plasmid. [4]

1) Origin of replication;

- to allow the plasmid to replicate in concert with the host DNA;
- GOI replicated along with vector

2) Selectable / Genetic Marker (2 markers involved)

- Usually an antibiotic resistant gene e.g. ampicillin resistance / Lac Z gene;
- 1st marker to select for bacterial cells which have taken in the plasmid / successfully transformed bacterial cells
- 2nd marker to select for recombinant bacterial cells / cells that have taken in the recombinant plasmid.
- Via replica plating / blue-white colonies selection ;

3) Multiple cloning/restriction sites

- found within the second selectable marker (e.g. tetracycline or Lac Z gene)
- allows insertion of GOI that treated with different R.E
- Insertion will disrupt the function of the selectable marker e.g tetracycline resistance gene or Lac Z gene that allows selection for recombinant plasmid.

(1/2 mark for each point)

(1 mark for labeled drawing)



(b) Using a named example, outline the process of cloning a eukaryotic gene into bacteria cells to produce eukaryotic protein in genetic engineering for therapeutic purposes. [10]

Example: (½ for name. ½ for function of protein)

- Insulin
- Treat diabetes patients by maintaining blood glucose levels.
- Human growth hormone
- To treat dwarfism
- Other valid example

1. Construction of recombinant DNA molecule (foreign gene + vector DNA)

a. Source of gene

mRNA for insulin

- mRNA **from** pancreas/ beta cells of pancreatic cells;
- **RT** is used to reverse transcribe mRNA to form complementary DNA (cDNA);
- **Alkaline/RNase H** to degrade the mRNA strand;
- **DNA pol** then synthesizes the second DNA strand;
- Mentioned use of linkers to create sticky ends (since cDNA will be blunt);
- Example of linkers;

b. Creation of recombinant DNA molecule using recombinant DNA technology

- Vector and insulin gene must be cleaved with same restriction enzyme at specific recognition sequence/site;
- So that complementary base pairing can form via hydrogen bonds between insulin gene fragment and vector;
- Sticky ends increases efficiency;
- Reason for use of vectors (unstable linear insulin gene/ OriR/ selectable marker gene);
- Example (of vector/ restriction enzyme);
- DNA ligase is added to seal phosphodiester bond;

2. Introduction of recombinant DNA molecule into living host (e.g. bacteria)

- Via transformation;
- Method of transformation;
- Reason of method (to create permeability/ transient enlargement of pores);
- Example of host cell;
- Purpose of living host (provide enzyme machinery);

3. Multiplication of recombinant DNA molecule ;

4. Cell growth/ division;



5. Screening/ Selection & identification of clones with recombinant DNA molecules

- Insertional inactivation;
- Lac Z/ Blue-white selection or antibiotic selection;
- Description of either lac Z or antibiotic selection (max 2 marks);

6. Purification of insulin product

- Upscale/ production in biofermenters with appropriate nutrient both/medium (max 1 mark);
- Extraction and purification of insulin protein from bacteria cells to be packaged.
- Post-translational modification/ combining the 2 chains via disulphide bonds;

(c) Discuss the goals and implications of the human genome project, including the benefits and difficult ethical concerns. [6]

Goals:

- To determine the sequence of the 3.2 billion base pairs that made up the human genome
- To identify all human genes (approximated 20 000-25 000 currently)
- To sequence the genome of model organisms
- Development and enhancement of resources for future research and commercial applications like DNA sequencing methods etc
- To explore gene function
- To study human variation like DNA polymorphisms
- To address the ethical, legal and social issues of genomic research through the implementation of the program (ELSI)

1 mark each,

Max for goals: 2 marks

Implications (Benefits):

Molecular Medicine

- Genes associated with certain diseases like breast cancer, cystic fibrosis and liver diseases have been pinpointed → development of diagnostic tests that can show the predisposition to these diseases.
- Understanding the genetic causes for diseases can in the long term lead to better and more targeted treatments and disease management.
- Creation of more effective and personalized drugs based on one's genetic profile.
- Creation of drugs based on molecular information.

Forensics

- Characterization of large DNA sequences and whole genome will be feasible when DNA sequencing technologies progress → more RFLP loci to be studied for more precise identification of individuals.



Bioarchaeology, Anthropology, Evolution, and Human Migration

- Comparing the human genome with other sequenced genomes of other organisms provide new insights into the evolutionary relationships.
- Study evolution through germline mutations in lineages
- study migration of different population groups based on maternal inheritance
- Study mutations on the Y chromosome to trace lineage and migration of males
- Compare breakpoints in the evolution of mutations with ages of populations and historical events

Advances in biological research

- Studies of genes are no longer limited to one or few genes at the same time. With the sequence of the whole genome available, research can be done on a wider scale like studying all the transcripts within a particular tissue and how different genes and proteins work together.
- Sharing of research information

1 mark each with elaboration

Max for benefits: 2 marks

Implications (Ethical issues):

- Ability to diagnose a genetic disorder before the disease sets in/before any treatment becomes available may do more harm than good. It creates risk of social stigmatism, anxiety and helplessness.
- Risk of genetic discrimination
 - Access of genetic information by insurance agencies. Is it considered appropriate for insurance agencies to access and use your genetic information? Now, they can check your family history of diseases and can increase premiums accordingly. So why can't they use genetic information in the same manner?
 - Could employers obtain and use genetic information against existing workers or to screen potential employees?
 - DNA fingerprinting as a personal identification may be useful, but it provides many intimate details about the person, e.g. susceptibility to certain genetic disorders, or the legitimacy of birth etc. This will increase the potential risks of discrimination by government, schools, employers etc.



- Commercialization of products including property rights (patents, copyrights, and trade secrets) and accessibility of data and materials.
 - Patenting gene fragments potentially hinders research and innovation of new technologies that involve the use of the patented research data due to the high cost involved.
 - Monopolization by certain biotech company that owns certain patents in the gene diagnostic market.
- Medical advancements that come with studying of the genome bring with them potential risks → gene therapy has caused the deaths of 6 patients during clinical trials.
- Potential threat **of eugenics** (the improvement of the human species by regulating heredity) → eliminating undesirable traits from the gene pool and alter these traits through genetic intervention.
- Implications of behavioral genetics (study of the genetics basis for behavior)
 - What social consequences would genetic diagnoses of such traits as intelligence, criminality, or homosexuality have on society? What effect would the discovery of a behavioral trait associated with increased criminal activity have on our legal system? If we find a "gay gene," will it mean greater or lesser tolerance? Will it lead to proposals that those affected by the "disorder" should undergo treatment to be "cured" and that measures should be taken to prevent the birth of other individuals so afflicted?

1 mark each for elaboration

Max for ethical issues: 2 marks



QUESTION SIX

- (a) Explain how ATP is synthesised at the mitochondrial inner membrane from 1 molecule of glucose for cellular activities, with reference to how the inner membrane is adapted for the synthesis.

1.	2 molecules of reduced NAD per glucose molecule are produced	1/2
2.	In cytoplasm during glycolysis	1/2
3.	And transported into the mitochondria by the mitochondrial shuttle	1/2
4.	8 molecules of reduced NAD (per glucose) / 4 molecules (per pyruvate that enters mitochondria)	1/2
5.	2 molecules of reduced FAD (per glucose)/ 1 molecule (per pyruvate that enters mitochondria)	1/2
6.	Synthesised in the matrix of mitochondria during Krebs cycle and Link reaction	1/2
7.	Reduced NAD and reduced FAD will enter the electron transport chain	1/2
8.	Located at the inner membrane of the mitochondria	1/2
9.	ETC comprises of a number of proteins embedded in the mitochondria	1/2
10.	With tightly bound/associated electron carriers	1/2
11.	The inner membrane is highly folded to form cristae to increase the surface area for reactions to take place	1/2
12.	At the ETC, a dehydrogenase enzyme catalyses the removal of H atoms/electrons from reduced NAD	1/2
13.	Electrons are from reduced NAD transferred from one electron carrier to another	1/2
14.	Electrons from reduced FAD is transferred directly to an electron carrier	1/2
15.	Oxygen is the final electron acceptor	1/2
16.	accepts electron and H ⁺ to form water	1/2
17.	When electrons are pass from one carrier to another of lower energy levels,	1/2
18.	energy is liberated	1/2
19.	Which is used to pump H ⁺ from the matrix into the intermembranal space	1/2
20.	thus creating a proton gradient/electrochemical gradient	1/2
21.	As the inner membranes are impermeable to protons	1/2
18.	H ⁺ which accumulates in the intermembranal space	1/2
19.	will diffuse back into the matrix down a proton/electrochemical gradient	1/2
20.	Through the ATPase	1/2
21.	H ⁺ diffuses down a concentration gradient	1/2
22.	H ⁺ is attracted to the less positively charged matrix	1/2
23.	Energy is released (as H ⁺ moves through ATPase)	1/2
24.	Energy is used to attach Pi to ADP to form ATP	1/2
25.	3 ATP is produced per reduced NAD that enters ETC	1/2
26.	2 ATP produced per reduced FAD that enters ETC	1/2
27.	A total of 34 ATP is produced per glucose that enters aerobic respiration.	1/2
28.	Which is transported out of the mitochondrial via transport proteins for cellular work	1/2
TOTAL:		14
MAXIMUM:		12



(b) Compare and contrast the Krebs cycle and the Calvin cycle.

1.	Both are cyclic – acceptor biomolecule is regenerated		1
2.	Both involve hydrogen carriers		1
3.	Occurs in matrix of mitochondria	Occurs in stroma of chloroplast	½, ½
4.	Occurs in all aerobically respiring cells	Occurs in plant cells/algae/blue-green bacteria	½, ½
5.	Catabolic reaction – breakdown of pyruvate	Anabolic reaction/ formation of triose phosphate	½, ½
6.	via series of oxidation processes	Via series of reduction processes	½, ½
7.	Can only occur when O ₂ is present	Does not require oxygen	½, ½
8.	Can occur in the absence of light	Indirectly requires light in order for Calvin cycle to proceed	½, ½
9.	Release of CO ₂	Usage of CO ₂	½, ½
10.	NAD and FAD as hydrogen carriers	NADP as hydrogen carriers	½, ½
11.	Production of ATP	Expenditure of ATP	
12.	Acetyl coenzyme A as the substrate that enters the cycle	CO ₂ as the substrate that enters the cycle	
13.	Material that is regenerated is oxaloacetate	Material that is regenerated is RuBP	½, ½
TOTAL:			13
MAXIMUM:			8

*If points are not corresponding mentioned (compared); no marks
Marks not given also if answers not in complete sentences*

